

From Lived Experiences to Legal Frameworks: Reconciling Public Perceptions with Formal Data for Context-Aware Accessibility Resources

1st Sanchita Kamath
School of Information Sciences
University of Illinois
Urbana-Champaign 61820
ssk11@illinois.edu

2nd Eryclis Silva
School of Information Sciences
University of Illinois
Urbana-Champaign 61820
ersilva2@illinois.edu

3rd Nitya Tirumalasetti
School of Information Sciences
University of Illinois
Urbana-Champaign 61820
nht2@illinois.edu

Abstract—This study aims to reconcile public perceptions with formal reporting to create a comprehensive resource that addresses the needs of dis/abled individuals in the domains of physical activity, mental health, and workplace accommodations. Quantitative insights were first derived from the National Health Interview Survey (NHIS) to identify trends in these areas, providing a foundational understanding of dis/abled individuals’ experiences. Qualitative insights were then obtained through a thematic analysis of Reddit discussions, capturing public perceptions and lived experiences to complement the statistical trends from NHIS data. A comparative analysis was conducted to explore how these two perspectives align and diverge, offering a nuanced understanding of dis/ability-related challenges. To translate these insights into an actionable resource, the sentiment analysis of Reddit data, conducted through multiple natural language processing (NLP) models, and legal accessibility frameworks such as the Americans with Disabilities Act (ADA), Section 504 were integrated into a context-aware system capable of providing accurate and empathetic guidance. While the deliverable is a functional large language model (LLM) system, the study’s focus remains on bridging the gap between statistical reporting and the lived experiences of dis/abled individuals to inform the development of accessible, accurate, and user-centric resources.

Keywords—accessibility, generative artificial intelligence, user-centric design, resource generation, mixed-methods research

I. INTRODUCTION

The intersection of physical activity, mental health, and workplace behavior for dis/abled individuals presents unique challenges that significantly impact their quality of life and inclusion in society. dis/abled individuals often face compounded difficulties in accessing physical activity opportunities, managing mental health, and navigating workplace environments. These barriers can exacerbate social isolation, limit physical engagement, and contribute to workplace exclusion.

A. Problem Statement

Public datasets like the National Health Interview Survey (NHIS) provide a broad overview of these issues, offering valuable insights into trends related to physical activity, mental health, and employment for dis/abled individuals. However, such data lacks the depth and personal context required to fully understand the daily experiences and emotional struggles

these individuals encounter. To bridge this gap, this research compares NHIS data with sentiment analysis of Reddit posts from dis/abled communities. Reddit offers a rich, personal perspective, capturing how individuals discuss their lived experiences related to workplace accessibility, physical activity, and mental health challenges. By analyzing both public data and online narratives, this study aims to answer key questions about how dis/abled individuals perceive these domains and how their experiences align or differ from survey data. Ultimately, this comparison will inform the development of a Large Language Model (LLM) capable of providing empathetic, context-aware responses to accessibility and mental health inquiries, improving technological support for dis/abled individuals. Through our work, we aim to answer the umbrella question of - How can insights from Reddit discussions about physical activity, mental health, and workplace behavior for dis/abled individuals be used to train a Large Language Model (LLM) to provide empathetic, context-aware responses? We hypothesize that - (a) Hypothesis 1: Sentiments expressed by dis/abled individuals on Reddit about physical activity, mental health, and workplace behavior will not differ from the trends observed in NHIS data; (b) Hypothesis 2: Integrating Reddit sentiments and legal accessibility data into LLM training will improve the LLM’s ability to provide empathetic and context-aware responses related to workplace behavior, mental health, and physical activity for dis/abled individuals. To verify our hypotheses we drafted some research questions.

B. Research Questions

- RQ1 What specific themes and sentiments related to physical activity, mental health, and workplace behavior are most frequently discussed by dis/abled individuals on Reddit?
- RQ2 To what extent do the sentiments expressed in Reddit posts align with the trends observed in NHIS data on physical activity, mental health, and workplace participation for dis/abled individuals?
- RQ3 How can the integration of Reddit sentiment and legal accessibility data improve the accuracy and empathy of an LLM’s responses to inquiries about workplace behavior,

mental health, and physical activity for dis/abled individuals?

II. RELATED WORK

Accessibility encompasses more than physical spaces, extending to mental health, workplace inclusion, and the socio-technical systems [1] that shape daily life for dis/abled individuals [2]. The authors aim to critically examine the interplay between accessibility, mental health, physical activity (PA), and the workplace [3], highlighting existing gaps in policy, practice, and research [4]–[7], and have thus conducted this literature review to understand work being done in the field.

A. Physical Activity and Disability

PA is a well-documented determinant of physical and mental health. It is particularly critical for dis/abled individuals, offering benefits such as reduced symptoms of anxiety and depression, improved mobility, and greater overall well-being [8]. However, numerous studies highlight the systemic barriers dis/abled populations face in accessing PA opportunities [9]. These include infrastructural inaccessibility in gyms, limited availability of adaptive equipment, and social stigma surrounding dis/ability in fitness spaces [10]. Researchers have identified a gap in tailoring PA programs to the specific needs of dis/abled individuals, calling for greater incorporation of universal design principles to address these challenges [11]. While interventions promoting PA for dis/abled individuals exist, they are often narrow in scope, focusing on one type of dis/ability or setting. Broader, community-focused approaches that integrate accessible fitness options into public health initiatives remain limited. Furthermore, there is little consensus on the most effective models for scaling such interventions, leaving a critical void in the literature on sustainable, inclusive PA initiatives.

B. Mental Health and Accessibility

Mental health challenges among dis/abled populations are both pervasive and underexplored. Dis/abled individuals experience disproportionately higher rates of depression and anxiety, often linked to societal exclusion, stigma, and the compounded stress of navigating inaccessible environments [12]. Research shows that mental health services frequently fail to accommodate dis/ability-specific needs, such as sensory-friendly spaces for neurodivergent individuals or mental health professionals trained in assistive communication methods for deaf individuals [13]. Online support communities, such as forums and social media platforms, have emerged as critical resources for dis/abled individuals seeking mental health support [14]. However, the literature on the efficacy and potential pitfalls of these digital spaces remains sparse. While some studies emphasize their role in fostering peer support and reducing isolation [15], [16], others caution against the risk of misinformation and the uneven access to digital tools among marginalized groups [17]. Expanding research on how these communities intersect with traditional mental health services could provide a more nuanced understanding of their benefits and limitations [18].

C. Workplace Inclusion and Accessibility

Workplace accessibility for dis/abled individuals has received significant attention in the fields of organizational behavior and dis/ability studies. Research consistently highlights the challenges dis/abled employees face, including discrimination, insufficient accommodations, and a lack of understanding of neurodiversity [19]. Policies such as the Americans with Disabilities Act (ADA) have aimed to address these issues, yet studies suggest a significant gap between policy intentions and practical implementation [20], [21]. Recent work has explored the potential of workplace interventions, such as assistive technologies and inclusive hiring practices, in improving outcomes for dis/abled workers [3], [22]. The emerging field of neurodiversity in the workplace highlights the unique contributions of neurodivergent employees [23] but underscores the need for long-term strategies that extend beyond recruitment to retention and career development [24].

D. Legal Frameworks and Their Role in Accessibility

Legal frameworks such as the ADA and Section 504 have been pivotal in advancing accessibility rights. However, literature critiques their limitations in addressing the evolving nature of accessibility needs, particularly in the digital era [25]. Comparative studies of international accessibility laws reveal significant variability in enforcement mechanisms and the scope of coverage. For example, while European countries emphasize the “design for all” approach [26], U.S. frameworks tend to focus on individualized accommodations [27]. This divergence suggests the need for cross-disciplinary studies to identify best practices and inform more comprehensive policy development. Additionally, integrating legal knowledge into technological systems, such as Natural Language Processing (NLP) tools [28], has shown potential in improving accessibility compliance by translating abstract provisions into actionable guidance [29].

III. METHODS

This study employs a mixed-methods approach to explore how dis/abled individuals perceive and experience physical activity, mental health, and workplace behavior by integrating data from NHIS and Reddit discussions. This dual-method design enables us to juxtapose the broader, quantitative trends from NHIS data with the qualitative, sentiment-rich insights from online communities. By comparing these two data sources, we aim to address how dis/abled individuals’ lived experiences, as shared online, align or differ from public data, and how these insights can inform the development of an empathetic, context-aware LLM.

For the quantitative aspect of the study, the NHIS dataset was utilized. This dataset provides a national snapshot of the health and employment status of dis/abled individuals in the United States, with specific variables related to physical activity, self-reported mental health, and employment outcomes. Data cleaning and preparation involved filtering for relevant dis/ability types—cognitive, mobility, auditory, and visual impairments—while focusing on key variables such as

the frequency of physical activity, self-assessment of mental health, and employment status. This dataset forms the baseline for identifying trends in physical health, mental health, and workplace participation among dis/abled individuals. Data was collected from Reddit, a widely-used online platform where individuals discuss a range of topics, including those relevant to dis/ability, health, and workplace behavior. Posts were mined from dis/ability-focused subreddits that cater to communities with cognitive, mobility, auditory, and visual impairments. Data collection focused on discussions over the past year and top posts, which were selected based on relevance and engagement. Given Reddit’s API limitation of retrieving 100 posts per query, we developed a dynamic batching system to gather a representative sample of discussions across these subreddits.

The posts were cleaned and deduplicated, and those without significant commentary or relevance to the study’s themes were excluded upon manual inspection. Sentiment analysis was applied to both the post titles and the associated comments. This analysis was conducted separately for titles and comments to capture the divergence between the often sensationalized sentiments in titles and the more reflective, detailed sentiments found in user comments. The sentiment analysis involved a combination of polarity scoring (positive, negative, or neutral) and deeper qualitative analysis to identify recurring themes related to physical activity, mental health, and workplace behavior. We adopted the RoBERTa model due to its ability to capture contextual and semantic nuances in both short and long texts, such as social media comments, providing higher accuracy in sentiment classification (positive, negative, and neutral). Simultaneously, another approach used was VADER, a rule-based technique known for efficiently capturing sentiment in short, informal texts like online posts. However, VADER exhibited a bias toward negative sentiment, often misclassifying neutral comments as negative. As a result, VADER was applied exclusively to post titles, as it demonstrated better performance in analyzing short, informal online texts, which closely resemble the structure of titles. This approach yielded more accurate sentiment classifications for titles compared to its inconsistent results with comments.

To leverage the strengths of both models, VADER sentiment analysis was used for titles, and RoBERTa sentiment analysis was applied to comments. The overall sentiment of each post was then evaluated using a weighted averaging approach, reflecting the relative importance of titles and comments in determining sentiment. Titles, serving as concise summaries, were assigned a weight of 40% while the more context-rich comments were given a weight of 60%. The aggregated sentiment score was calculated as a weighted combination of the two models’ outputs, ensuring a balanced representation of both components. The combined sentiment score was subsequently mapped into three categories (positive, neutral, or negative) based on predefined thresholds. This process enabled a comprehensive and interpretable sentiment classification, effectively balancing the insights provided by both the titles and the associated comments. Besides, to explore

the thematic structure within the Reddit data, we performed topic modeling using advanced natural language processing techniques. Sentence embeddings were generated using a pre-trained Sentence Transformer, capturing the semantic richness of the text. To manage the high-dimensional nature of these embeddings, we applied dimensionality reduction with UMAP, which effectively preserves both local and global relationships in lower-dimensional spaces. Subsequently, HDBSCAN was employed to cluster the reduced embeddings into coherent groups, enabling the identification of topics. To refine the extracted topics and generate diverse, contextually accurate labels, we employed Maximal Marginal Relevance (MMR) to balance relevance with keyword diversity, complemented by OpenAI’s GPT-3.5-turbo for precise and meaningful labeling.

This combined method of analysis—quantitative trends from NHIS data and qualitative insights from Reddit—allows for a comprehensive examination of how dis/abled individuals experience and perceive physical activity, mental health, and workplace behavior. It also provides the foundation for training an LLM that can offer empathetic, context-aware responses informed by both statistical data and personal narratives. The tone of the sentiment is imported into the LLM’s system prompt to provide a context-aware response by matching Reddit titles to the user comment/input provided to the LLM. In addition to the sentiment analysis and topic modeling, we also collected legal data on accessibility regulations, including information such as the law’s name, year, type, summary, code, application, and link. This legal dataset is also imported into the LLM system response, ensuring it can provide accurate and legally grounded responses to inquiries related to workplace accessibility and other legal aspects of dis/ability.

IV. SOLUTION

Our prototype involves the development and testing of the LLM designed to deliver empathetic and context-aware responses tailored to the needs of dis/abled individuals. Please review Fig. 1 to understand the pipeline of our system. The LLM incorporates a range of accessibility features to ensure it serves individuals with diverse needs, including those with cognitive, mobility, auditory, and sensory impairments. These features include **text-to-speech functionality**, which provides spoken responses to make the system accessible to users who cannot rely on visual text, and **auditory processing cues**, such as a beep sound loop, which signal that the system is actively processing a request. These cues are particularly important for users who may not receive visual confirmation of system activity, ensuring inclusivity in the interaction experience.

The model also integrates natural language processing capabilities that simplify complex legal and technical text, making it easier for users to comprehend responses related to workplace accommodations, accessibility regulations, and mental health or physical activity recommendations. By including sentiment analysis of personal narratives from Reddit and legal data, the LLM can provide responses that are both factually accurate and empathetically tuned to the lived experiences of dis/abled individuals. This dual capability ensures the

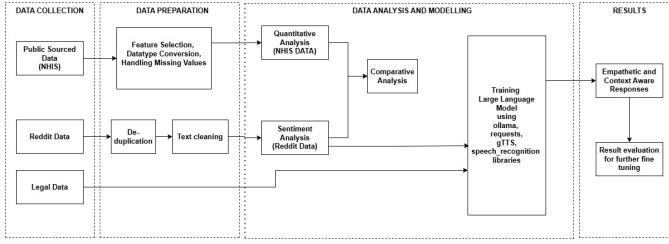


Fig. 1. Prototype Solution

model can address both informational and emotional needs effectively. In addition to these accessibility considerations, the LLM integrates legal data on accessibility regulations. By grounding responses in real-world laws and policies, the LLM empowers dis/abled individuals to advocate for their rights and navigate complex situations more effectively. The legal data is processed alongside the sentiment analysis data, allowing the model to address user inquiries from both an emotional and informational perspective.

To evaluate the system, a beta test was conducted with an ultra-low-vision (ULV) individual with total blindness in their left eye and a visual acuity of 20/2000. This testing phase aims to assess the accessibility, usability, and responsiveness of the LLM. Although the beta test focuses on the experience of a ULV individual, the accessibility features are designed to benefit a much broader audience of dis/abled users. The test will examine how effectively the auditory features, speech recognition, and text-to-speech outputs work in practice, as well as how the model handles nuanced, context-sensitive inquiries. The inclusion of a ULV tester in the beta phase provides valuable insights into how the system performs under real-world conditions for users with significant vision impairments. This approach ensures the LLM is not only functional but also truly accessible. Accessibility features such as auditory cues and clear spoken responses are critical for creating an inclusive system, as they allow users to engage with the model without relying on vision or other senses that may be impaired. The beta test will also provide feedback on the model's ability to deliver concise yet meaningful responses, balancing empathy with efficiency to meet the needs of its users. The results of the beta test informed an iterative refinement process, where feedback is used to enhance the model's performance and accessibility, aligning responses with users' lived experiences. The solution aims to bridge the gap between data-driven insights and the personal narratives of dis/abled individuals, creating a system that is both inclusive and impactful. You may find our collective code on GitHub at SK-143381/is-597-hcdis. This repository includes the code to our LLM system, our quantitative analysis code as well as all our cleaned data.

V. RESULTS AND EVALUATION

A. Quantitative Analysis

In Table V-A, PA shows a positive Pearson correlation with both depression (0.388) and anxiety (0.456), suggesting a

moderate relationship between increased PA and heightened mental health symptoms. Although this result contrasts with the common understanding of PA as a mental health intervention, it is consistent with the findings from the regression analysis in Table V-A and Table V-A, where PA_activity does not significantly affect the severity of either depression or anxiety ($p = 0.813$ and $p = 0.732$, respectively). In both models, the effect size for PA is small, reinforcing the idea that in this particular population, physical activity may not have the expected protective effect on mental health. This finding warrants further exploration, as it may suggest that PA interventions, though effective in general populations, might need to be adapted to be more effective for dis/abled individuals.

TABLE I
BASIC DESCRIPTIVE STATISTICS

Correlation Type	PA and Depression	PA and Anxiety	PA and Tiredness	PA and Days Absent	Anxiety and Tiredness	Depression and Tiredness
Pearson Correlation	0.388	0.456	-0.003	0.000	0.140	0.165
Spearman Correlation	0.059	0.109	-0.045	0.012	0.140	0.174

The regression analysis for depression in Table V-A shows that hours worked and income have minimal and statistically insignificant effects ($p = 0.229$ and $p = 0.599$, respectively). This aligns with the near-zero Pearson correlation between PA and days absent (0.000016) found in Table V-A, suggesting that workplace-related factors like time worked or income are not strong predictors of depression severity in this sample. However, level of tiredness shows a statistically significant relationship with depression ($p = 0.035$), with higher levels of tiredness linked to increased depression severity. This finding is echoed in Table V-A, where tiredness and depression have a Pearson correlation of 0.165, highlighting fatigue as a critical factor in mental health. Given this significant correlation and regression result, the data suggest that fatigue may be a more relevant intervention point than physical activity for addressing depressive symptoms in dis/abled populations.

TABLE II
REGRESSION ANALYSIS: DEPRESSION SEVERITY

Variable	Estimate	Std. Error	t-value	$\Pr(> t)$
(Intercept)	2.656	0.344	7.720	7.59×10^{-14}
PA_activity	0.009	0.037	0.237	0.813
Hours worked per week	-0.004	0.003	-1.202	0.230
Income	0.044	0.084	0.525	0.600
Tiredness	0.133	0.063	2.112	0.035
Age	-0.022	0.003	-7.012	8.60×10^{-12}
Sex	0.175	0.099	1.767	0.078

The regression results for anxiety in Table V-A largely mirror those for depression. PA again shows a statistically

insignificant effect ($p = 0.732$), and workplace factors such as hours worked and income remain non-significant predictors ($p = 0.305$ and $p = 0.644$, respectively). Interestingly, level of tiredness is positively related to anxiety, as shown in Table V-A, where the Pearson correlation between anxiety and tiredness is 0.140. However, in the regression analysis, this relationship is not statistically significant ($p = 0.373$), which may suggest that while tiredness is associated with anxiety, it does not independently predict anxiety severity when controlling for other variables. The Spearman correlation for tiredness and anxiety in Table V-A (0.140) supports this moderate association but does not suggest a strong relationship.

TABLE III
REGRESSION ANALYSIS: ANXIETY SEVERITY

Variable	Estimate	Std. Error	t-value	Pr(χ^2 —t—)
(Intercept)	2.671	0.361	7.407	6.41×10^{-13}
PA_activity	0.013	0.039	0.341	0.
Hours worked per week	-0.003	0.003	-0.974	0.331
Income	0.040	0.088	-0.461	0.645
Tiredness	0.059	0.066	0.890	0.374
Age	-0.020	0.003	-6.268	8.56×10^{-10}
Sex	0.156	0.104	1.503	0.134

One of the most consistent findings across all tables is the strong relationship between age and mental health outcomes. Both regression models for depression and anxiety (Tables V-A and V-A) show that younger individuals experience higher levels of both depression and anxiety, with age being a statistically significant predictor in both cases ($p \leq 0.001$ for depression and $p \leq 0.001$ for anxiety). This relationship is not explicitly highlighted in Table V-A, but the regression results strongly indicate that age is a major factor influencing mental health severity, more so than PA or workplace-related variables. Younger individuals may be more vulnerable to mental health issues, possibly due to greater stress, life transitions, or other unmeasured factors.

PA has no significant effect on MH, contrary to popular belief and academic literature. We attribute this difference to three possible explanations - (1) people did not answer accurately, misinterpreted questions or did not recall severity well; (2) the demographic has not been well studied enough to make assumptions (3) sampled individuals who participated in the survey are outliers and not representative of the population (which is more unlikely). In the case of dis/abled individuals, Figure 2 suggests that those who report more physical activity might also be experiencing more depression and anxiety, possibly due to the additional effort and barriers they face in engaging in PA. Interestingly, there is a near-zero correlation between PA and days absent from work (0.00002), indicating that physical activity levels do not seem to impact the number of workdays missed by dis/abled individuals. Similarly, workplace factors like hours worked per week and income show no significant correlation with mental health variables, suggesting

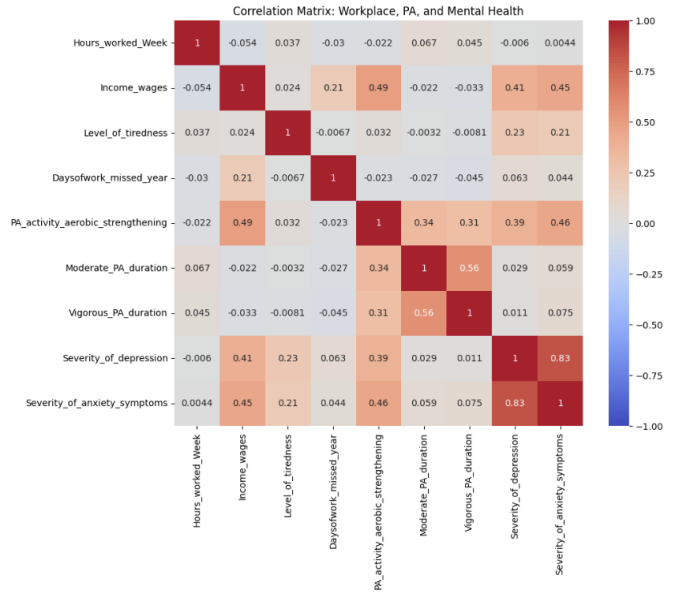


Fig. 2. Correlation Matrix for relevant variables

that these are not major contributors to depression or anxiety in this population. One notable finding is the relationship between tiredness and depression (0.16), which, though moderate, is statistically significant. This suggests that higher levels of tiredness are associated with more severe depression among dis/abled individuals. The correlation between anxiety and tiredness (0.14) also points to a link between workplace fatigue and mental health, though it is weaker. The matrix highlights that factors like tiredness play a more significant role in mental health outcomes for dis/abled individuals than expected, while physical activity, despite its widely acknowledged benefits, may require more tailored approaches to be effective for this population.

B. Qualitative Analysis

The rationale behind the topic modeling and sentiment analysis is to provide a structured understanding of the diverse conversations within Reddit communities, capturing nuanced themes and sentiments related to physical activity, mental health, and workplace behavior among dis/abled individuals. By identifying 24 distinct topics available in Table IV, the analysis highlights both the challenges faced and the coping strategies employed by this population, offering valuable insights into their everyday realities.

One prominent theme that emerges is the difficulty faced by deaf individuals in accessing mental health services (Topic 1). Many users express frustration over the scarcity of professionals fluent in sign language, which limits their ability to receive adequate psychological support. This leads to a sense of isolation when seeking mental health care, with individuals reporting feeling marginalized within the broader healthcare system. This lack of accessible services is a significant barrier for many within the deaf community, underscoring the need

Topic ID	Topic Name	Description
1	Deaf Mental Health Services Oppression	Barriers in accessing mental health care, emphasizing systemic issues for the deaf community.
2	Mental Health dis/ability Validity	Recognition and legitimacy of mental health dis/abilities in societal contexts.
3	Dis/ability Advocacy and Communication	Efforts to improve communication accessibility for dis/abled individuals.
4	Blind Friendly Exercise Activities	Adaptive exercise routines for individuals with visual impairments.
5	Braille for Independent Work	Role of Braille literacy in enhancing independence and career opportunities.
6	Neurodiversity and Workplace Inclusion	Advocacy for inclusive workplaces for neurodivergent individuals.
7	Social Security Dis/ability Programs	Navigation of benefits like SSI/SSDI and associated barriers.
8	Navigating Life with Visual Impairment	Emotional resilience and community support for visually impaired individuals.
9	Coping with Heartbreak and Loss	Experiences of dis/abled individuals in overcoming relationship challenges.
10	Struggles of Living with Dis/abilities	Economic hardships and societal stigma faced by dis/abled individuals.
11	Living with Hearing Loss	Accessibility challenges and social isolation due to hearing impairments.
12	Mental Health Treatment Challenges	Obstacles in accessing effective mental health care for dis/abled individuals.
13	Hypotonia and Hypermobility EDS	Experiences of living with hypotonia and hypermobile Ehlers-Danlos syndrome.
14	Emotional Struggle with Vision Diagnosis	Coping with psychological impact of vision-related diagnoses.
15	Mobility Challenges in Social Settings	Navigation issues in public spaces due to mobility limitations.
16	Workplace and School Abuse	Reports of abuse and discrimination in workplaces and educational institutions.
17	Challenges of Living with Dis/abilities	Broad challenges including financial strain and public misunderstandings.
18	Mental Health Crisis and Jails	Treatment of mentally dis/abled individuals within the justice system.
19	Accessible Public Transportation Options	Advocacy for improved public transit accessibility.
20	Vision and Mobility Assistance in India	Accessibility challenges and solutions specific to India.
21	End-of-Life Autonomy and Mental Health	Autonomy in end-of-life decisions and related mental health support.
22	Blind Life Skills Training	Training programs for independence and life skills for blind individuals.
23	Mental Health Hospital Access Discrimination	Discrimination in accessing mental health care facilities.
24	Medical Foster Care Abuse Reporting	Addressing abuse in medical foster care systems.

TABLE IV
SUMMARY OF TOPICS IDENTIFIED IN REDDIT DISCUSSIONS

for policies that prioritize training mental health professionals to meet the specific needs of these populations.

In parallel, there is an evident frustration in discussions surrounding mental dis/abilities and the misunderstanding of their conditions (Topic 2). Users often share experiences where their struggles are dismissed or overlooked due to the invisibility of their dis/abilities. This theme highlights a broader societal issue: the need for greater awareness and understanding of mental health conditions and dis/abilities that may not be immediately apparent to others. As participants navigate the complexities of mental health care, they frequently encounter barriers that exacerbate their feelings of marginalization and misjudgment. Inclusive language and dis/ability advocacy (Topic 3) also surface as key concerns, with debates over terminology reflecting the diverse ways in which dis/ability is understood and represented. These conversations expose internal conflicts within the dis/ability community itself, with some advocating for specific terms such as “person with a disability” while others resist these labels. Despite these differences, there is a shared desire for recognition and respect. The theme of inclusion extends beyond language to encompass practical actions aimed at fostering a more inclusive environment in both societal and workplace settings. A significant portion of the discussion focuses on the role of adapted physical activities for individuals with visual impairments (Topic 7), with participants expressing empowerment through their participation in sports like goal-ball. These activities are not only seen as a means to improve physical health but also as a way to enhance mental well-being. However, while the enthusiasm for these sports is palpable, there is also an acknowledgment of the challenges faced in

accessing suitable facilities, pointing to the need for more inclusive spaces that can support people with dis/abilities in pursuing recreational activities.

The importance of Braille and other assistive technologies for visually impaired individuals (Topic 7) is another central theme. Many users share their frustrations with devices that do not accommodate Braille or other accessibility needs, which impedes their independence and autonomy. The sentiment here is one of resilience, as individuals continue to seek out solutions and alternatives that enable them to navigate the world more effectively, despite the limitations they encounter in technology and design. The workplace is another domain where inclusion is a significant concern (Topic 8). Discussions surrounding neurodiversity emphasize the strength that can be found in diverse cognitive approaches, yet there remains a notable gap in the workplace when it comes to embracing this diversity. While advocacy for neurodivergent individuals is strong, many individuals continue to face barriers such as slow institutional change and discrimination. The discussion reveals a disconnect between the recognition of the value of neurodivergent employees and the actual steps taken to create inclusive environments that accommodate their needs. The complexity of navigating dis/ability benefits (Topic 9) also stands out as a recurring theme. Many users discuss their struggles with accessing social security programs, which are often complicated by state-specific requirements and extensive documentation. This complexity is a significant barrier for individuals trying to secure the benefits they need. The sentiments expressed here are a mix of frustration with bureaucratic hurdles and a desire for more streamlined processes that would make it easier for individuals with dis/abilities to access critical

resources.

Alongside these structural challenges, many individuals share stories of emotional resilience and the importance of community support (Topic 4). These themes are particularly evident in discussions around visual impairments, where users highlight the role of supportive communities in overcoming the emotional and social challenges associated with vision loss. Despite the emotional hardships, there is a common thread of empowerment, with individuals finding strength in their ability to learn new skills and build connections with others facing similar challenges. Finally, financial struggles and the emotional toll of living with a dis/ability (Topic 6) are often discussed, particularly in relation to inadequate dis/ability payments and the stigmatization of invisible illnesses. Many participants express feelings of rejection and frustration, not only with the financial support they receive but also with the societal misunderstandings that perpetuate negative stereotypes about dis/abilities. This speaks to the urgent need for economic reform and broader public education campaigns to reduce stigma and improve access to both financial support and mental health services.

C. Correlation between NHIS and Reddit Data

Across the Reddit dataset, negative sentiment dominated (67.46%), particularly in discussions on workplace inclusion and mental health, where it reached 72.22% under workplace-related themes. This pattern reflects systemic frustrations with discrimination, inadequate accommodations, and the psychological toll of navigating inaccessible environments. In contrast, positive sentiment—though limited (26.59%)—emerged in discussions about community support and successful adaptations, such as the use of Braille for independence and inclusive physical activities. A comparative analysis of NHIS data and Reddit narratives reveals substantial alignments and divergences in addressing the challenges faced by dis/abled individuals across domains like mental health, workplace inclusion, and physical activity. Both datasets converge on the prominence of mental health and workplace challenges. NHIS data provides a quantitative perspective, capturing variables such as mental health levels, life satisfaction, and workplace participation. Reddit narratives similarly highlight barriers to effective accommodations and the psychological effects of inaccessible environments. The high prevalence of negative sentiment in Reddit posts, particularly in workplace and mental health discussions, aligns with NHIS findings on systemic issues such as low income levels, reduced working hours, and high absenteeism rates. These parallels underscore the urgent need for interventions targeting workplace inclusion and mental health support.

Reddit discussions further enrich these insights by detailing specific challenges that echo broader NHIS trends. For instance, themes like the difficulty finding interpreters and lack of workplace understanding for neurodivergent employees mirror NHIS indicators of diminished workplace participation and its mental health implications. Similarly, barriers to mental health care raised in Reddit narratives—such as inaccessible

resources for the deaf community—reflect NHIS findings on poor general health and low life satisfaction, emphasizing the critical need for tailored mental health interventions. Both datasets also highlight the impact of physical activity and mobility challenges on mental health. However, they acknowledge that physical activity alone offers limited benefits without the integration of tailored, accessible solutions. The analysis shows a clear difference between the general trends found in NHIS data and the specific, real-life experiences shared on Reddit. While NHIS offers broad, quantitative insights, Reddit provides detailed stories that reflect the day-to-day challenges of dis/abled individuals. For instance, Reddit discussions about mental health barriers for the deaf community and workplace abuse offer emotional depth that NHIS data doesn't capture. These findings highlight the need to combine both quantitative and qualitative data to fully understand the intersection of dis/ability, mental health, and workplace inclusion.

Reddit posts describe daily life issues like the lack of mental health professionals who know sign language and the emotional impact of being left out of workplace social activities. These personal stories uncover the complex realities that dis/abled individuals face—something NHIS data alone cannot show. Additionally, Reddit brings attention to problems that NHIS data overlooks or doesn't cover in detail. Discussions about workplace and school abuse, discrimination in mental health care, and foster care abuse (Topics 16, 23, 24) reveal the systemic and personal struggles that NHIS metrics miss. Reddit narratives also explore the emotional side of dis/ability. Stories about coping with heartbreak and loss (Topic 9), dealing with vision issues (Topic 14), and the weight of social stigma give a deeper understanding of mental health challenges. NHIS may count these issues, but it doesn't explain the emotional burden behind them. Another key difference is how each source talks about the role of physical activity. NHIS data suggests that physical activity helps improve mental health and workplace outcomes. However, Reddit users often challenge this idea. While physical activity has potential benefits, these users point out that without accessible solutions, it's not always realistic. Issues like fatigue and inaccessible infrastructure make it hard for many to stay active. These personal accounts highlight a gap in NHIS data: the need for practical, tailored solutions that meet real-world needs.

D. User Evaluation

The user evaluation focused on assessing the functionality, clarity, and user-centricity of the system through two separate chat interactions. Each interaction was designed to address distinct user needs, allowing for a holistic qualitative evaluation of the system's performance.

1) *Chat 1: Physical Activity Recommendations:* The first chat explored the system's ability to provide personalized physical activity recommendations suitable for cold weather. The user initiated the interaction with the query: "Hi, can you give me some physical activities I can do and it's cold outside." The system responded by suggesting a variety of indoor exercises, such as indoor walking, chair yoga, resis-

tance band exercises, and Tai Chi. The recommendations were concise, practical, and catered to accessibility, as evidenced by the system's inclusion of modifications like seated yoga and low-impact options for individuals with limited mobility and reduced visual acuity. The user followed up with a more focused query about resistance band training, asking for "two things." The system provided detailed instructions for bicep curls and tricep extensions, including information on sets, repetitions, and safety considerations.

Subsequently, the user asked: "Where can I install my resistance bands and how?" The system delivered actionable suggestions, including using door anchors or stability balls, accompanied by safety tips like ensuring secure anchoring and adequate space.

2) *Chat 2: Legal and Workplace Accommodation Support:* The second chat evaluated the system's capacity to provide legally informed advice about workplace accommodations. The user inquired about the legal obligation to receive Braille materials, stating: "My employer has given me an appointment to do however for me to complete it I require that the materials be in Braille. Am I legally obligated to receive said materials in Braille?" The system cited the Americans with Disabilities Act (ADA) and Section 504 of the Rehabilitation Act, explaining the employer's obligation to provide reasonable accommodations unless doing so posed an undue hardship. When the user requested a draft email to request accommodations, the system provided a well-structured template.

The user then requested: "I think one thing that would be helpful to have in the email is a set of steps or resources that the employer could refer to in creating those Braille materials." The system responded by including action steps and suggesting organizations such as the National Federation of the Blind (NFB) and the American Printing House for the Blind (APH).

3) *User Observations and Feedback:* Following are some of the features that the user liked. They said that the system demonstrated exceptional capability in generating practical and relevant recommendations, effectively addressing the user's queries in both chats. Its knowledge of accessibility laws was another highlight. The system also excelled at maintaining coherence and relevance in follow-up questions, providing clear, actionable instructions, and generating structured email templates that reflected the user's specific needs. These strengths underscored its potential to support dis/abled users effectively.

However, the participant also noticed that the system exhibited several shortcomings that impacted their experience. Responses were often "too long," which led to user fatigue and detracted from the conversational flow. Formatting issues, such as spaces between text, disrupted readability and made responses appear less polished. In Chat 2, the tone of the email draft was critiqued as "doesn't read like a sub-ordinate reaching out," reflecting an overly stern and authoritative style inappropriate for the user's context. Additionally, while the system mentioned valuable resources like the National Federation of the Blind, it failed to include URLs or actionable links, a notable gap given the user's request for resources that

the employer could use to create Braille materials.

The participant provided targeted suggestions for improvement based on their interaction with the system. They noted that the text-to-speech component was "too slow," with a request to "increase speed of lady" for better engagement. Responses were also described as "little straight out of the gate," indicating a lack of smooth conversational entry in some cases. Nonetheless, the system was praised for being "good at understanding what accommodations are available," highlighting its strength in accurately identifying and addressing accessibility-related needs. Overall, while the system's foundational functionality was acknowledged, the feedback emphasized the need for iterative refinements to enhance usability and user satisfaction.

VI. DISCUSSION

We address RQ1 through a detailed analysis of qualitative data extracted from Reddit discussions. This analysis reveals several key themes that reflect the lived experiences of individuals with dis/abilities, particularly in the realms of physical activity, mental health, and workplace behavior. These themes provide both insight into the challenges faced by these individuals and the coping strategies they employ to navigate societal barriers. The most frequently discussed themes related to physical activity and mental health involve both the barriers faced and the coping strategies used. Regarding physical activity, topics related to goal-ball and other adapted activities for visually impaired individuals (Topics 4 and 8) demonstrate a strong desire for inclusion and participation in sports, with sentiments of empowerment and enthusiasm. However, there is also a clear lack of accessible spaces to practice these activities, which limits many individuals' access to such experiences. This theme reflects the importance of promoting accessible physical activities and developing assistive technologies to maximize the participation of individuals with dis/abilities in sports and recreational activities. In terms of mental health, Topic 1 and Topic 2 highlight the significant challenges faced by deaf individuals and those with mental dis/abilities in accessing appropriate psychological support services. The frustration and isolation are predominant in many accounts, with the lack of trained professionals to meet the specific needs of these populations. This points to a large gap in the provision of inclusive and accessible mental health services, emphasizing the urgent need for policies that promote the training of specialized therapists and the development of more inclusive support systems for individuals with dis/abilities. Workplace inclusion was widely discussed in Topic 6, focusing on the barriers still faced by neurodivergent individuals in the professional environment. While there is significant advocacy for creating more inclusive environments, with a strong sense of advocacy, there is also an acknowledgment that change is slow, and many individuals still face discrimination and a lack of understanding about the capabilities of neurodivergent individuals. This reflects the need for quicker political and practical changes to ensure true workplace inclusion, highlighting the importance of inclusion

policies and awareness practices in companies. Finally, Topics 7, 8, 9 and 10 discuss the challenges related to navigating benefits (Topic 7), emotional and social barriers faced by visually impaired individuals (Topic 8), and the financial and emotional struggles of individuals with dis/abilities (Topics 9 and 10). These topics underline critical issues of inequity in accessing resources and support, as well as the economic barriers that limit the quality of life for individuals with dis/abilities.

We address RQ2 by analyzing the correlation between NHIS data and Reddit discussions to understand how quantitative findings align with the lived experiences of dis/abled individuals. This combined approach reveals both convergence and divergence in themes related to mental health, workplace inclusion, and physical activity. The NHIS data provides a quantitative foundation, identifying systemic challenges such as low income, high absenteeism, and reduced workplace participation. This aligns with the predominantly negative sentiment found in Reddit discussions, which reflect frustrations over discrimination, lack of accommodations, and the psychological toll of navigating inaccessible environments. Notably, workplace-related themes exhibited even higher negative sentiment, underscoring persistent barriers to inclusion. Reddit narratives enrich these quantitative insights by offering detailed, emotional accounts that NHIS data cannot capture. In terms of physical activity, while NHIS data suggests its benefits for mental health and workplace outcomes, Reddit discussions challenge the feasibility of these benefits without accessible infrastructure and adaptive solutions. Users express difficulties stemming from fatigue and environmental barriers, revealing a gap in NHIS data that underscores the need for practical, context-specific solutions. Ultimately, integrating NHIS data with Reddit narratives offers a holistic understanding of dis/ability-related challenges. This dual approach emphasizes that while quantitative data identifies broad systemic issues, qualitative insights from Reddit illuminate the emotional and practical realities, guiding the development of more inclusive and targeted interventions.

We address RQ3 through the user evaluation of our system, which examines how effectively the integration of Reddit sentiment and legal accessibility data enables the system to deliver accurate, empathetic, and context-aware responses. The evaluation revealed that the combined use of these data sources allowed the system to provide responses that were informed by both legal frameworks and lived experiences, creating a balance between factual precision and relatability. The integration of legal data proved essential for ensuring the accuracy and reliability of the system's responses. By referencing established legal provisions and frameworks, the system was able to provide structured advice that aligns with users' rights and obligations. This demonstrates the system's capacity to transform abstract legal concepts into actionable guidance. However, the evaluation highlighted limitations in operationalizing this knowledge, such as the omission of direct links or resources, which could have enhanced the usability of the responses by providing users with tangible

next steps. The use of Reddit sentiment data contributed to the system's ability to address the emotional and contextual dimensions of user queries. By incorporating insights from community discussions, the system attempted to align its responses with the lived experiences of dis/abled individuals, fostering empathy. Nevertheless, the evaluation identified gaps in how sentiment data was applied, particularly in adapting tone dynamically to fit relational contexts. While sentiment-informed training improved contextual awareness, it lacked the refinement necessary to consistently match the conversational tone to the user's intent, such as adjusting for professionalism in workplace-related queries.

VII. CHALLENGES AND LIMITATIONS

The study is constrained by its reliance on specific data sources, which introduces biases that may not fully represent the diversity of dis/abled individuals' experiences. The Reddit sentiment data, while rich in personal narratives, predominantly reflects the perspectives of users active on social media platforms. This reliance on a single online community risks excluding insights from dis/abled individuals who lack access to or choose not to engage with these forums, leading to potential gaps in addressing their lived experiences. Additionally, the NHIS data, used to contextualize trends, captures only predefined variables that may not align with the granular or emergent themes from Reddit discussions. This misalignment restricts the depth of insights derived from comparing these datasets.

Problems in the comparison of the publicly sources NHIS data and the Reddit data:

1. Cognitive Decline has been considered as "difficulty remembering or concentrating" which is very different from the medical definition of the term as well as perspectives observed in the Reddit Data.
2. Variables to measure workplace issues include 'number of hours worked', 'income', 'level of tiredness', 'last time worked', 'main reason not working' (which is a set of predefined options from which the individuals have chosen), 'absence', 'days of work missed per year'. These variables are hard to compare to the social perspectives of data collected from Reddit.

A significant limitation lies in the scope of legal data integration. Since we created this dataset without any legal backing or knowledge, we run the risk of misrepresenting information being relayed to the user. Further, while the study included accessibility regulations, its focus was limited to overarching U.S.-based frameworks such as the ADA. This excludes regional, industry-specific, or international legal variations that could influence the advice provided by the system. Consequently, the system is unable to address users operating in non-U.S. contexts or those subject to unique organizational policies, thereby narrowing its applicability and relevance.

The study's evaluation methodology also presents challenges. Conducting user testing with only two distinct chats limits the generalizability of the findings, as it does not capture the range of potential scenarios users might encounter.

Furthermore, the reliance on a single user for evaluation reduces the robustness of feedback, as individual biases or experiences may overemphasize certain issues while overlooking others. This raises questions about the representativeness of the insights gained and whether the findings can be extrapolated to broader populations of dis/abled users. Another limitation is the absence of longitudinal analysis. The study lacks follow-up testing to assess how repeated interactions with the system influence user behavior or whether the system's recommendations lead to sustained improvements in accessibility outcomes. Without this data, it is unclear whether the system has a long-term impact on reducing barriers or improving inclusion for dis/abled individuals. Finally, the study does not account for the evolving nature of accessibility challenges and regulatory updates. Accessibility needs and legal requirements are dynamic, influenced by technological advancements, cultural shifts, and policy changes. By relying on static datasets, the study inherently struggles to address these temporal factors, which could affect the relevance and accuracy of both the training data and the system's outputs over time. These limitations suggest that future iterations of the study must incorporate more diverse data sources, broader user testing, and mechanisms for adapting to evolving accessibility contexts.

REFERENCES

- [1] K. E. Ringland, E. Carstensdottir, M. Tung, B. Seetharaman, T. Eagle, L. B. Baltaxe-Admony, and K. Weatherwax, "Together in the Mikrokosmos: Exploring Disabled Embodied Experiences in Designed Sociotechnical Worlds," *Proc. ACM Hum.-Comput. Interact.*, vol. 8, pp. 435:1–435:39, Nov. 2024.
- [2] V. Kameswaran, *Help Facilitates Accessibility: Understanding the Social and Technology-mediated Experiences of People with Visual Impairments in India*. Thesis, 2024. Accepted: 2024-05-22T17:25:38Z.
- [3] M. V. Gauci, *Enabling Technology in the Workplace: Exploring the dis/ability assemblage*. phd, University of Leeds, Sept. 2018.
- [4] P. Gelius, S. Messing, L. Goodwin, D. Schow, and K. Abu-Omar, "What are effective policies for promoting physical activity? A systematic review of reviews," *Preventive Medicine Reports*, vol. 18, p. 101095, June 2020.
- [5] G. W. Heath and D. Levine, "Physical Activity and Public Health among People with Disabilities: Research Gaps and Recommendations," *International Journal of Environmental Research and Public Health*, vol. 19, p. 10436, Jan. 2022. Number: 16 Publisher: Multidisciplinary Digital Publishing Institute.
- [6] A. Khalid and J. Syed, "Mental health and well-being at work: A systematic review of literature and directions for future research," *Human Resource Management Review*, vol. 34, p. 100998, Mar. 2024.
- [7] E. McAuley and K. S. Morris, "State of the Art Review: Advances in Physical Activity and Mental Health: Quality of Life," *American Journal of Lifestyle Medicine*, vol. 1, pp. 389–396, Sept. 2007. Publisher: SAGE Publications.
- [8] A. Mahindru, P. Patil, and V. Agrawal, "Role of Physical Activity on Mental Health and Well-Being: A Review," *Cureus*, vol. 15, no. 1, p. e33475, 2023.
- [9] K. Arora and G. Wolbring, "Kinesiology, Physical Activity, Physical Education, and Sports through an Equity/Equality, Diversity, and Inclusion (EDI) Lens: A Scoping Review," *Sports*, vol. 10, p. 55, Apr. 2022. Number: 4 Publisher: Multidisciplinary Digital Publishing Institute.
- [10] J. S. Lewis, "Understanding Barriers Preventing Those with Limited Mobility from Obtaining Equal Access and Opportunity to Exercise and Achieve Overall Health," in *Advances in Exercise and Health for People With Mobility Limitations* (D. Hollar, ed.), pp. 157–189, Cham: Springer International Publishing, 2019.
- [11] M. Geoffrey, D. Larry, J., and P. Patricia, *ACSM's Exercise Management for Persons With Chronic Diseases and Disabilities*, 4E. Human Kinetics, Mar. 2016. Google-Books-ID: mEjZCwAAQBAJ.
- [12] G. Brewer, *Disability in Higher Education: Investigating Identity, Stigma and Disclosure amongst Academics*. McGraw-Hill Education (UK), Apr. 2022. Google-Books-ID: Gx1tEAAAQBAJ.
- [13] M. Kersten, N. J. Wilson, A. Pracilio, V. Howie, J. Trollor, T. Buckley, J. Morphet, J. Bryce, K. Griffin, and A. Cashin, "A scoping review to inform an auditing framework evaluating healthcare environments for inclusion of people with intellectual disability and/or autism," *Journal of Intellectual Disabilities*, p. 17446295231174282, May 2023. Publisher: SAGE Publications Ltd.
- [14] M. Hall, "Disability, discourse and desire: Analyzing online talk by people with disabilities," *Sexualities*, vol. 21, pp. 379–392, Mar. 2018. Publisher: SAGE Publications Ltd.
- [15] B. G. Bell and J. Clegg, "An Ecological Approach to Reducing the Social Isolation of People With an Intellectual Disability," *Ecological Psychology*, vol. 24, pp. 159–177, Apr. 2012. Publisher: Routledge _eprint: <https://doi.org/10.1080/10407413.2012.673983>.
- [16] M. Chakraborti, M. Gitimoghaddam, W. H. McKellin, A. R. Miller, and J.-P. Collet, "Understanding the Implications of Peer Support for Families of Children With Neurodevelopmental and Intellectual Disabilities: A Scoping Review," *Frontiers in Public Health*, vol. 9, Nov. 2021. Publisher: Frontiers.
- [17] J. Yu, D. A. Bekerian, and C. Osback, "Navigating the Digital Landscape: Challenges and Barriers to Effective Information Use on the Internet," *Encyclopedia*, vol. 4, pp. 1665–1680, Dec. 2024. Number: 4 Publisher: Multidisciplinary Digital Publishing Institute.
- [18] J. S. Seng, W. D. Lopez, M. Sperlich, L. Hamama, and C. D. Reed Meldrum, "Marginalized identities, discrimination burden, and mental health: Empirical exploration of an interpersonal-level approach to modeling intersectionality," *Social Science & Medicine*, vol. 75, pp. 2437–2445, Dec. 2012.
- [19] D. C. Beardmore, "Exploring the Experiences of Dis/Abled STEM Graduate Students - ProQuest," 2023.
- [20] S. P. Harris, R. Gould, and G. Fujiura, "Enhancing rigor and practice of scoping reviews in social policy research: Considerations from a worked example on the Americans with Disabilities Act," *Work*, vol. 50, pp. 323–334, Jan. 2015. Publisher: IOS Press.
- [21] J. E. West, V. L. McLaughlin, K. G. Shepherd, and R. Cokley, "The Americans with Disabilities Act and the Individuals with Disabilities Education Act: Intersection, Divergence, and the Path Forward," *Journal of Disability Policy Studies*, vol. 34, pp. 224–234, Dec. 2023. Publisher: SAGE Publications Inc.
- [22] S. Welch-Grenier, *Breaking the Barrier of Employer Biases: Improving Employment Outcomes Through Employer-tailored Training*. PhD thesis, The Ohio State University, 2024.
- [23] B. Wen, H. van Rensburg, S. O'Neill, and T. Attwood, "Autism and neurodiversity in the workplace: A scoping review of key trends, employer roles, interventions and supports," *Journal of Vocational Rehabilitation*, vol. 60, pp. 121–140, Dec. 2023. Publisher: SAGE Publications.
- [24] S. Bölte, J. A. Carpini, M. H. Black, A. Toomingas, F. Jansson, P. B. Marschik, S. Girdler, and M. Jonsson, "Career Guidance and Employment Issues for Neurodivergent Individuals: A Scoping Review and Stakeholder Consultation," *Human Resource Management*, vol. n/a, no. n/a, 2024. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/hrm.22259>.
- [25] J. de Perio Wittman, "Moving Beyond the Basics of the ADA and Section 504: Opportunities for Equitable and Inclusive Access to Law Libraries, Collections, and Services," *Legal Reference Services Quarterly*, vol. 42, pp. 272–298, Oct. 2023. Publisher: Routledge _eprint: <https://doi.org/10.1080/0270319X.2023.2264690>.
- [26] K. Bendixen and M. Benktzon, "Design for All in Scandinavia – A strong concept," *Applied Ergonomics*, vol. 46, pp. 248–257, Jan. 2015.
- [27] R. Colker, "The Americans with Disabilities Act's Unreasonable Focus on the Individual The Disability Frame: Opportunities, Costs, and Constraints in the Broad Struggle for Inclusion," *University of Pennsylvania Law Review*, vol. 170, no. 7, pp. 1813–1850, 2021.
- [28] I. Mariani, M. Karimi, G. Concilio, G. Rizzo, and A. Benincasa, "Improving Public Services Accessibility Through Natural Language Processing: Challenges, Opportunities and Obstacles," in *Intelligent Systems and Applications* (K. Arai, ed.), (Cham), pp. 272–289, Springer International Publishing, 2023.

- [29] R. K. Behera, P. K. Bala, N. P. Rana, and Z. Irani, "Responsible natural language processing: A principlist framework for social benefits," *Technological Forecasting and Social Change*, vol. 188, p. 122306, Mar. 2023.